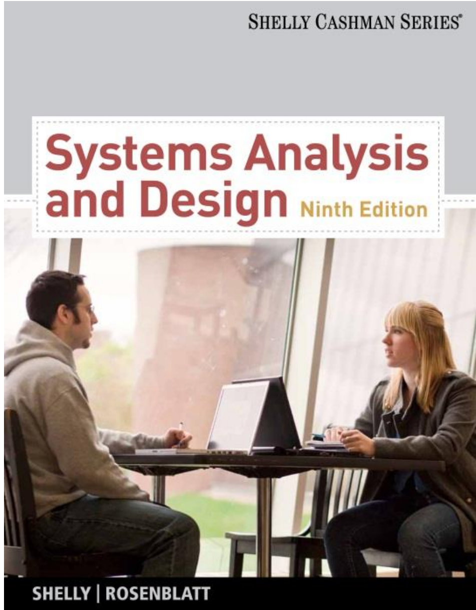
Systems Analysis and Design
9th Edition

Chapter 4

Requirements Modeling



Phase Description

- Systems analysis is the second of five phases in the systems development life cycle (SDLC)
- Will use requirements modeling, data and process modeling, and object modeling techniques to represent the new system
- Will consider various development strategies for the new system, and plan for the transition to systems design tasks

Chapter Objectives

- Describe systems analysis phase activities
- Explain joint application development (JAD), rapid application development (RAD), and agile methods
- Use a functional decomposition diagram (FDD) to model business functions and processes

Chapter Objectives

- Describe the Unified Modeling Language (UML) and examples of UML diagrams
- List and describe system requirements, including outputs, inputs, processes, performance, and controls
- Explain the concept of scalability

Chapter Objectives

- Use fact-finding techniques, including interviews, documentation review, observation, questionnaires, sampling, and research
- Define total cost of ownership (TCO)
- Conduct a successful interview
- Develop effective documentation methods to use during systems development

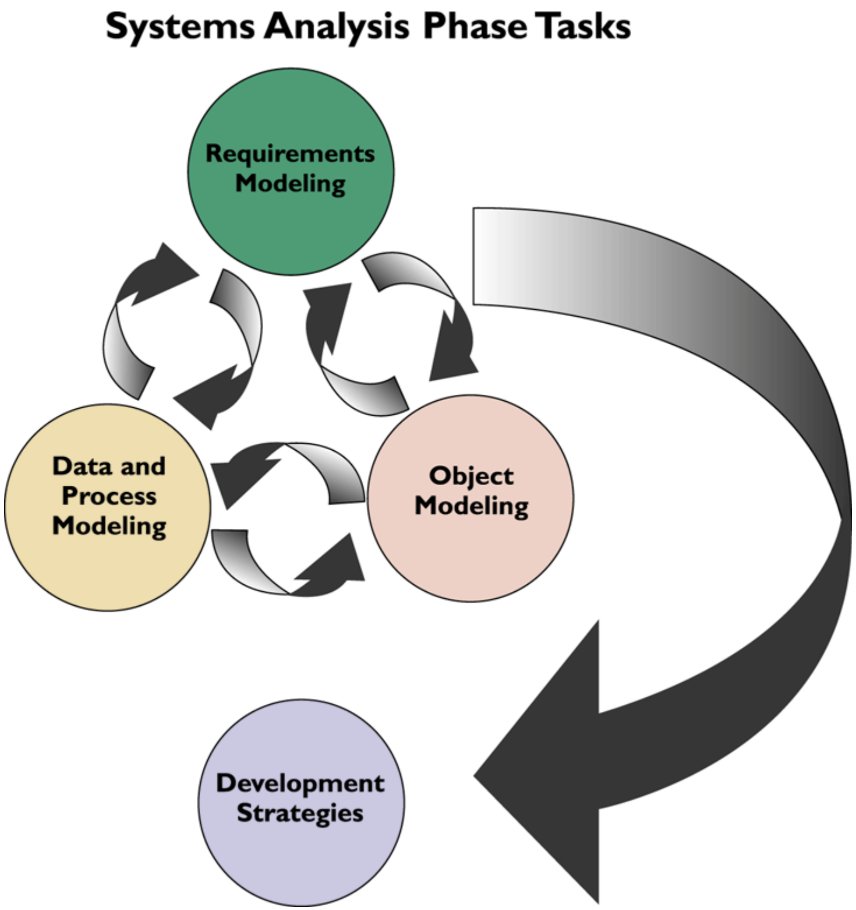
Introduction

- This chapter describes requirements modeling techniques and team-based methods that systems analysts use to visualize and document new systems
- The chapter then discusses system requirements and fact-finding techniques, which include interviewing, documentation review, observation, surveys and questionnaires, sampling, and research

Systems Analysis Phase Overview

- The overall objective of the systems analysis phase is to understand the proposed project, ensure that it will support business requirements, and build a solid foundation for system development
- You use models and other documentation tools to visualize and describe the proposed system

Systems Analysis Phase Overview



- Systems Analysis Activities
  - Requirements modeling
    - Outputs
    - Inputs
    - Processes
    - Performance
    - Security

Systems Analysis Phase Overview

- Systems Analysis Activities
  - Data and process modeling
  - Object Modeling
  - Development Strategies
    - System requirements document

Systems Analysis Phase Overview

- Systems Analysis Skills
  - Analytical skills
  - Interpersonal skills
- Team-Oriented Methods and Techniques
  - Joint application development (JAD)
  - Rapid application development (RAD)
  - Agile methods

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RAD

- http://www.cs.fsu.edu/~baker/swe2/restricted/notes/dennis/ch01.ppt
- **https://tinyurl.com/y2wqckdg**
- http://prachya-kmutnb.blogspot.com/2009/02/sa-assignment-4.html

Joint Application Development

- User Involvement
  - Users have a vital stake in an information system and they should participate fully
  - Successful systems must be user-oriented, and users need to be involved
  - One popular strategy for user involvement is a JAD team approach

Joint Application Development

- JAD Participants and Roles

JAD PARTICIPANT	ROLE
JAD project leader	Develops an agenda, acts as a facilitator, and leads the JAD session
Top management	Provides enterprise-level authorization and support for the project
Managers	Provide department-level support for the project and understanding of how the project must support business functions and requirements
Users	Provide operational-level input on current operations, desired changes, input and output requirements, user interface issues, and how the project will support day-to-day tasks
Systems analysts and other IT staff members	Provide technical assistance and resources for JAD team members on issues such as security, backup, hardware, software, and network capability
Recorder	Documents results of JAD sessions and works with systems analysts to build system models and develop CASE tool documentation

Joint Application Development

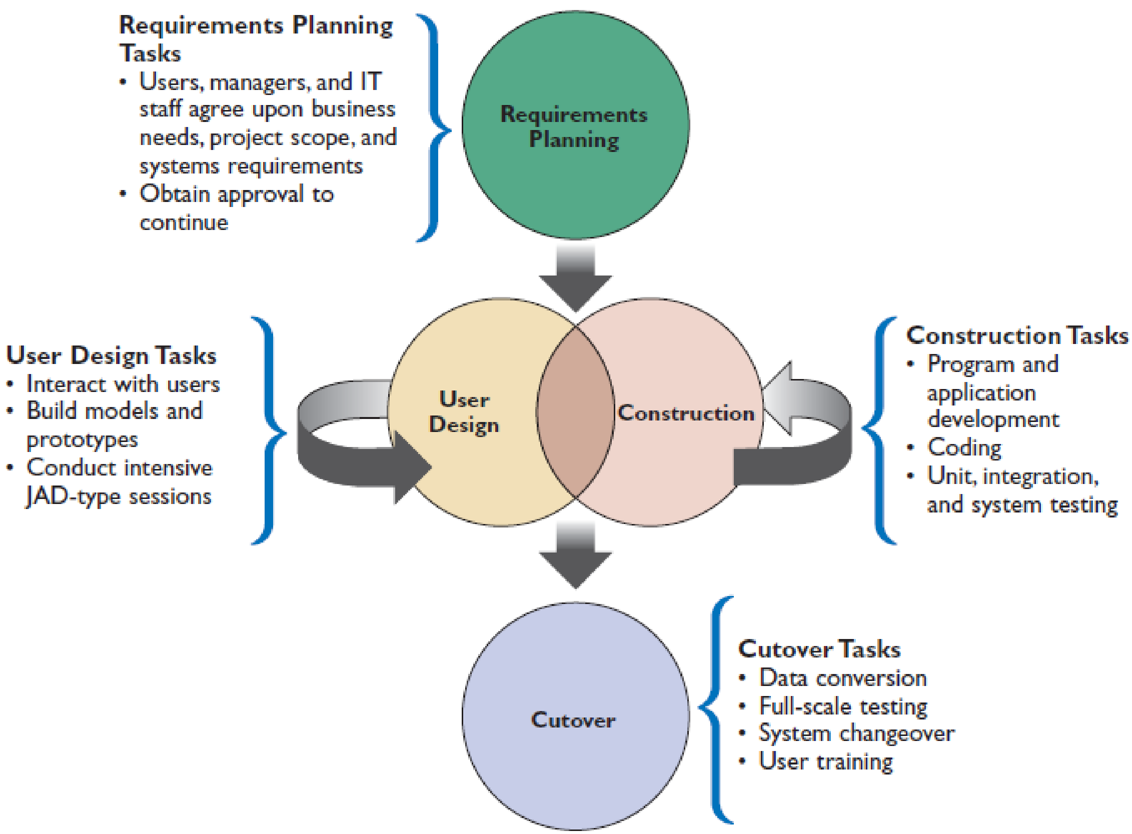
- JAD Advantages and Disadvantages
  - More expensive and can be cumbersome if the group is too large relative to the size of the project
  - Allows key users to participate effectively
  - When properly used, JAD can result in a more accurate statement of system requirements, a better understanding of common goals, and a stronger commitment to the success of the new system

Rapid Application Development

- Is a team-based technique that speeds up information systems development and produces a functioning information system
- Relies heavily on prototyping and user involvement
- Interactive process continues until the system is completely developed and users are satisfied

Rapid Application Development

- RAD Phases and Activities



Rapid Application Development

- RAD Objectives
  - To cut development time and expense by involving the users in every phase of systems development
  - Successful RAD team must have IT resources, skills, and management support
  - Helps a development team design a system that requires a highly interactive or complex user interface

Rapid Application Development

- RAD Advantages and Disadvantages
  - Systems can be developed more quickly with significant cost savings
  - RAD stresses the mechanics of the system itself and does not emphasize the company's strategic business needs
  - Might allow less time to develop quality, consistency, and design standards

Agile Methods

- Attempt to develop a system incrementally
- Agilian modeling toolset includes support for many modeling tools
- Some agile developers prefer not to use CASE tools at all, and rely instead on whiteboard displays and arrangements of movable sticky notes

Agile Methods

- Scrum is a rugby term
- Pigs include the product owner, the facilitator, and the development team; while the chickens include users, other stakeholders, and managers
- Scrum sessions have specific guidelines that emphasize time blocks, interaction, and team-based activities that result in deliverable software

Agile Methods

- Agile Method Advantages and Disadvantages
  - Are very flexible and efficient in dealing with change
  - Frequent deliverables constantly validate the project and reduce risk
  - Team members need a high level of technical and interpersonal skills
  - May be subject to significant change in scope

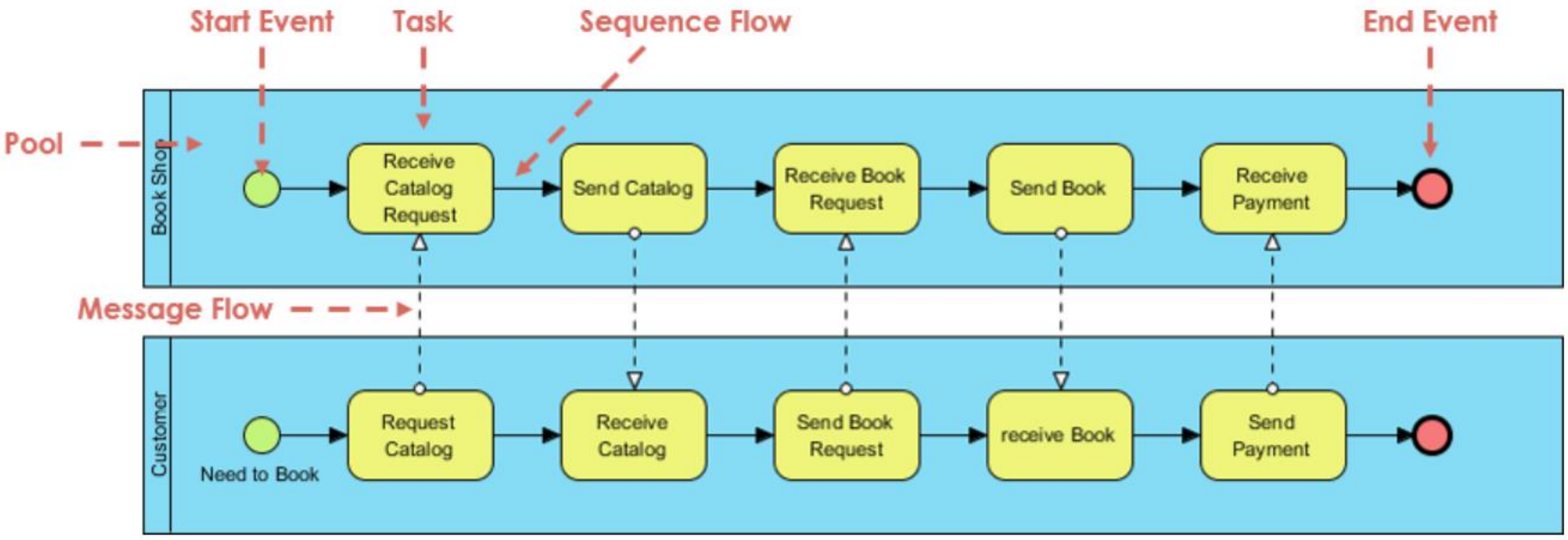
Modeling Tools and Techniques

- Involves graphical methods and nontechnical language that represent the system at various stages of development
- Can use various tools
- Functional Decomposition Diagrams
  - Functional decomposition diagram (FDD)
  - Model business functions and show how they are organized into lower-level processes

Modeling Tools and Techniques

- Business Process Modeling
  - Business process model (BPM)
  - Business process modeling notation (BPMN)
  - Pool
  - Swim lanes

Pools and Swimlanes



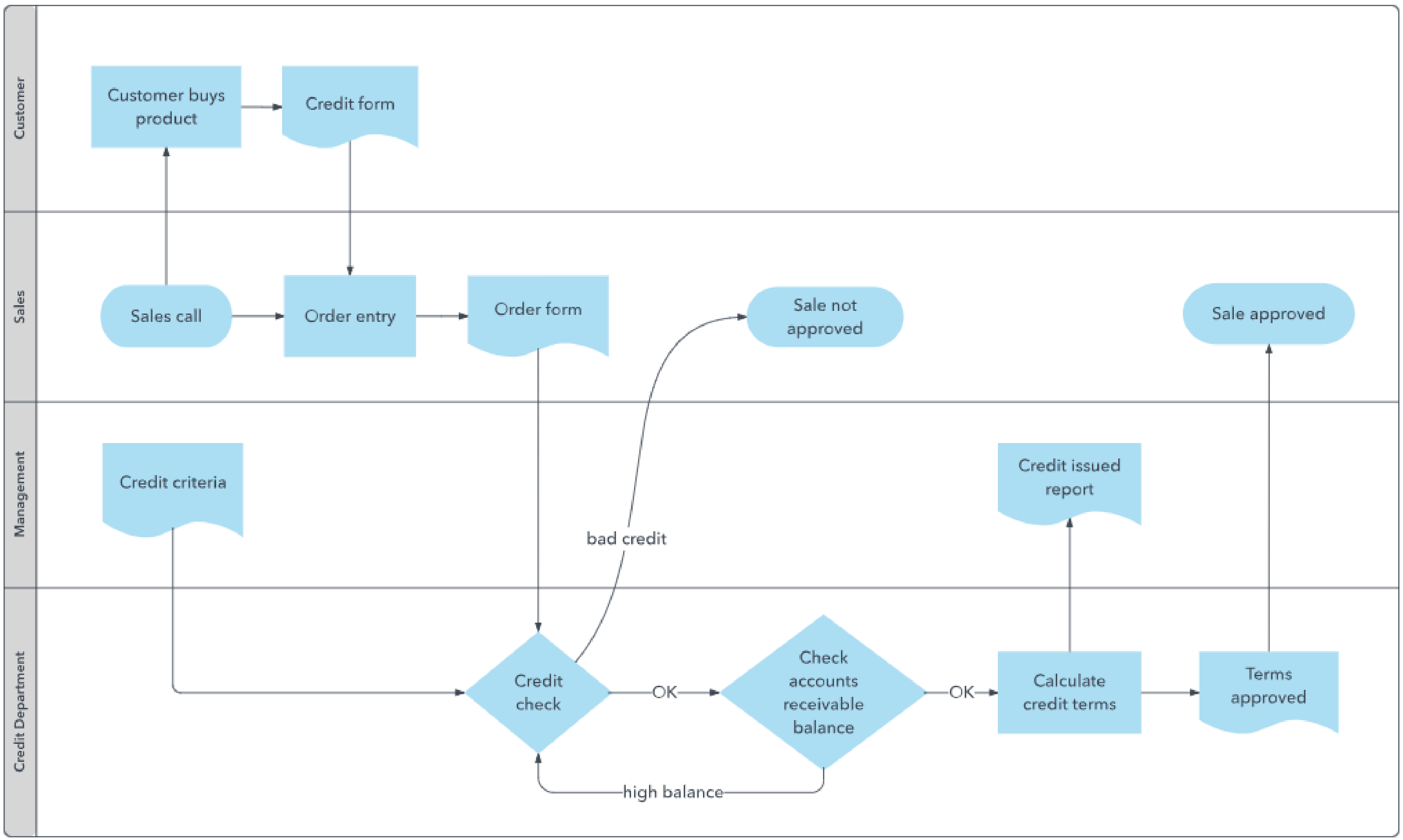
https://circle.visual-paradigm.com/pools-and-swimlanes/

Pools and Swimlanes

- Businesses are often divided logically into functional departments (eg. accounting, marketing, sales, warehouse etc). Swimlanes show each of these departments relevant to a process. Using swimlanes it can be seen when a process is crossing from the responsibility of one department to another. If two or more organisations are involved in a transaction (and typically for our purposes two are) then each organisation ay have its own internal swimlanes (i.e departments).

- Pool – represents a participant in a process. Usually in the context of B2B situations.
- Lane – represents a sub-partition within a pool. Lanes are used to organise and categorise activities.

Swim lanes



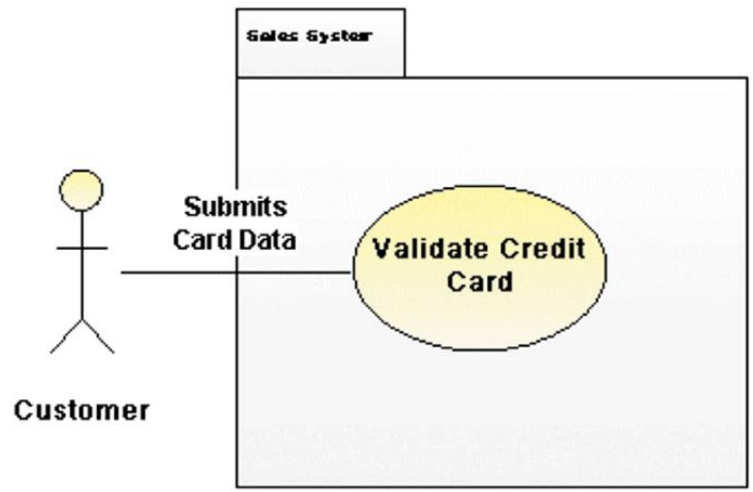
Customer
Sales
Management
Credit Department
Customer buys product
Credit form
Sales call
Order entry
Order form
Sale not approved
Sale approved
Credit criteria
Credit issued report
bad credit
Credit check
OK
Check accounts receivable balance
OK
Calculate credit terms
Terms approved
high balance

Modeling Tools and Techniques

- Data Flow Diagrams
  - Data flow diagram (DFD)
  - show how the system stores, processes, and transforms data
  - Additional levels of information and detail are depicted in other, related DFDs

Modeling Tools and Techniques

- Unified Modeling Language
  - Widely used method of visualizing and documenting software systems design
  - Use case diagrams
    - Actor
  - Sequence diagrams



System Requirements Checklist

- Outputs
  - The Web site must report online volume statistics every four hours, and hourly during peak periods
  - The inventory system must produce a daily report showing the part number, description, quantity on hand, quantity allocated, quantity available, and unit cost of all sorted by part number

System Requirements Checklist

- Inputs
  - Manufacturing employees must swipe their ID cards into online data collection terminals that record labor costs and calculate production efficiency
  - The department head must enter overtime hours on a separate screen

System Requirements Checklist

- Processes
  - The student records system must calculate the GPA at the end of each semester
  - As the final step in year-end processing, the payroll system must update employee salaries, bonuses, and benefits and produce tax data required by the IRS

System Requirements Checklist

- Performance
  - The system must support 25 users online simultaneously
  - Response time must not exceed four seconds

System Requirements Checklist

- Controls
  - The system must provide logon security at the operating system level and at the application level
  - An employee record must be added, changed, or deleted only by a member of the human resources department

Future Growth, Costs, and Benefits

- Scalability
  - A scalable system offers a better return on the initial investment
  - To evaluate scalability, you need information about projected future volume for all outputs, inputs, and processes

Future Growth, Costs, and Benefits

- Total Cost of Ownership
  - Total cost of ownership (TCO) is especially important if the development team is evaluating several alternatives
  - One problem is that cost estimates tend to understate indirect costs
  - Rapid Economic Justification (REJ)

Total cost of ownership (TCO) ต้นทุนการเป็นเจ้าของ

https://th.itpedia.nl/2017/07/30/totale-eigendomskosten-tco/

เพิ่มเติม SAAS

- https://tinyurl.com/2p8y7zkh

Fact-Finding

- Fact-Finding Overview
  - First, you must identify the information you need
  - Develop a fact-finding plan
- Who, What, Where, When, How, and Why?
  - Difference between asking what is being done and what could or should be done

Fact-Finding

- The Zachman Framework
  - Zachman Framework for Enterprise Architecture
  - Helps managers and users understand the model and assures that overall business goals translate into successful IT projects

Enterprise Architecture

https://www.dga.or.th/wp-content/uploads/2015/06/file_66935e266b45254af0398334d02e86e4.pdf

Interviews

- Step 1: Determine the People to Interview
  - Informal structures
- Step 2: Establish Objectives for the Interview
  - Determine the general areas to be discussed
  - List the facts you want to gather

Interviews

- Step 3: Develop Interview Questions
  - Creating a standard list of interview questions helps to keep you on track and avoid unnecessary tangents
  - Avoid leading questions
  - Open-ended questions
  - Closed-ended questions
  - Range-of-response questions

Interviews

- Step 4: Prepare for the Interview
  - Careful preparation is essential because an interview is an important meeting and not just a casual chat
  - Limit the interview to no more than one hour
  - Send a list of topics
  - Ask the interviewee to have samples available

Interviews

- Step 5: Conduct the Interview
  - Develop a specific plan for the meeting
  - Begin by introducing yourself, describing the project, and explaining your interview objectives
  - Engaged listening
  - Allow the person enough time to think about the question
  - After an interview, you should summarize the session and seek a confirmation

Interviews

- Step 6: Document the Interview
  - Note taking should be kept to a minimum
  - After conducting the interview, you must record the information quickly
  - After the interview, send memo to the interviewee expressing your appreciation
  - Note date, time, location, purpose of the interview, and the main points you discussed so the interviewee has a written summary and can offer additions or corrections

Interviews

- Step 7: Evaluate the Interview
  - In addition to recording the facts obtained in an interview, try to identify any possible biases
- Unsuccessful Interviews
  - No matter how well you prepare for interviews, some are not successful

การสัมภาษณ์ (INTERVIEW) + เครื่องมือการเก็บข้อมูล

http://aurapinjwy.blogspot.com/2014/10/interview.html

Other Fact-Finding Techniques

- Document Review
- Observation
  - Seeing the system in action gives you additional perspective and a better understanding of the system procedures
  - Plan your observations in advance
  - Hawthorne Effect

Hawthorne Effects

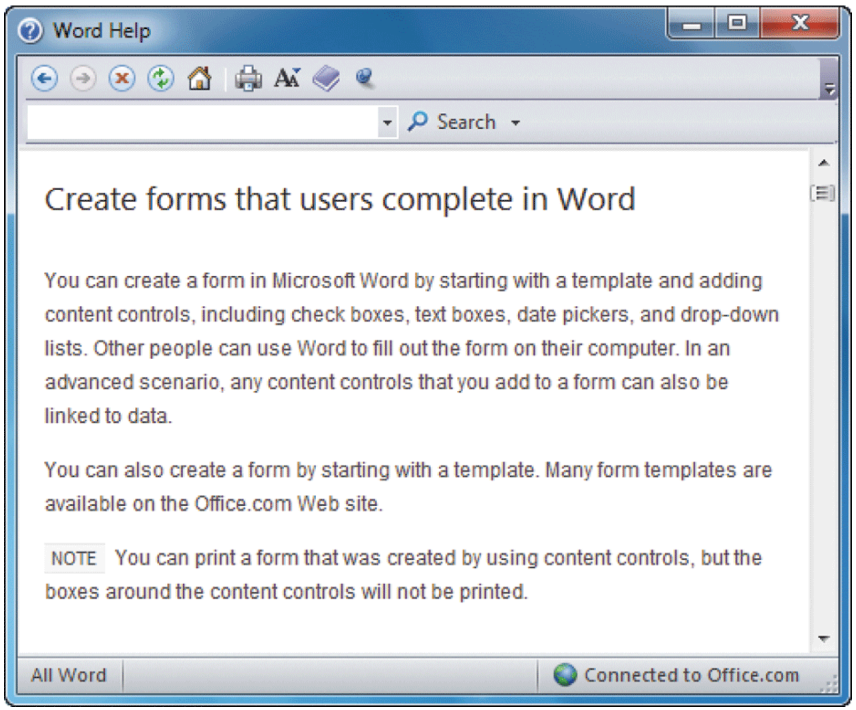
http://varanyu.blogspot.com/2009/06/hawthorne-effects.html

https://tinyurl.com/mvwtn324

***6M -> หนึ่งใน M คือ Morale ***

https://tinyurl.com/25y432md

Other Fact-Finding Techniques



- Questionnaires and Surveys
  - When designing a questionnaire, the most important rule of all is to make sure that your questions collect the right data in a form that you can use to further your fact-finding
  - Fill-in form

Other Fact-Finding Techniques

- Sampling
  - Systematic sample
  - Stratified sample
  - Random sample
  - Main objective of a sample is to ensure that it represents the overall population accurately

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http://pioneer.netserv.chula.ac.th/~jaimorn/re6.htm

Other Fact-Finding Techniques

- Research
  - Can include the Internet, IT magazines, and books to obtain background information, technical material, and news about industry trends and developments
  - Site visit

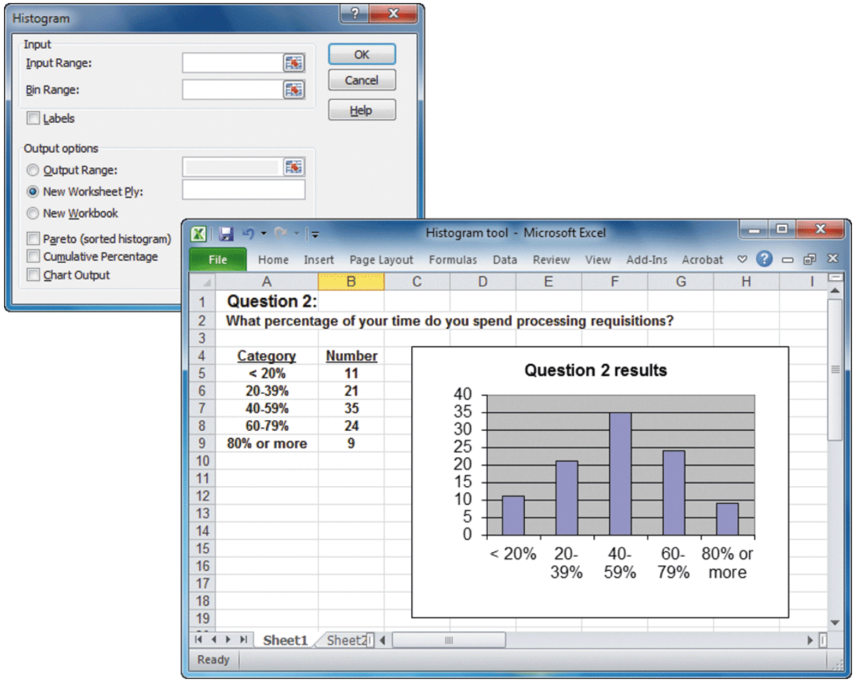
Other Fact-Finding Techniques

- Interviews versus Questionnaires
  - Interview is more familiar and personal
  - Questionnaire gives many people the opportunity to provide input and suggestions
  - Brainstorming
  - Structured brainstorming
  - Unstructured brainstorming

Documentation

- The Need for Recording the Facts
  - Record information as soon as you obtain it
  - Use the simplest recording method
  - Record your findings in such a way that they can be understood by someone else
  - Organize your documentation so related material is located easily

Documentation



- Software Tools
  - CASE Tools
  - Productivity Software
    - Word processing, spreadsheets, database management, presentation graphics, and collaborative software programs
    - Histogram

Documentation

- Software Tools
  - Graphics modeling software
  - Personal information managers
  - Wireless communication devices

Preview of Logical Modeling

- At the conclusion of requirements modeling, systems developers should have a clear understanding of business processes and system requirements
- The next step is to construct a logical model of the system
- IT professionals have differing views about systems development methodologies, and no universally accepted approach exists

Chapter Summary

- The systems analysis phase includes three activities: requirements modeling, data and process modeling, and consideration of development strategies
- The main objective is to understand the proposed project, ensure that it will support business requirements, and build a solid foundation for the systems design phase

Chapter Summary

- The fact-finding process includes interviewing, document review, observation, questionnaires, sampling, and research
- Systems analysts should carefully record and document factual information as it is collected, and various software tools can help an analyst visualize and describe an information system